# Supporting Information: Reliability of Short Molecular Dynamics Runs as Screening Labels

Francisco Hernando-Gallego,*,† Dimitar Vasilev-Vasilev,‡ Diego García-Santos,‡ Alfonso Campo-Camacho,¶ and Alberto Gómez-Gómez¶

†*Department of Applied Mathematics, Escuela de Ingeniería Informática de Segovia, Universidad de Valladolid, 40005 Segovia, Spain*

‡*Department of Computer Science, Escuela de Ingeniería Informática de Segovia, Universidad de Valladolid, 40005 Segovia, Spain*

¶*Independent Researcher, Spain*

E-mail: fhernando@uva.es

## Data set

Table S1: Data used in this study. All systems: LiTFSI in an amorphous polymer, PCFF+, 353 K.

| | 5 ns runs | 50 ns runs |
|---|---|---|
| Trajectories | 6,137 | 133 |
| MSD curves used ($Li^+$, TFSI) | 100 lags each | 100 lags each |
| Longest lag in the curve | 5 ns | 50 ns |
| Systems also simulated at the other length | 133 | 133 |
| Systems simulated twice at this length | 94 | n/a |

Raw trajectories of the 133 long runs were read as a stream, keeping only ion positions,

the solvating polymer atoms and the system centre of mass. The origin-averaged MSD computed from them reproduces the curves distributed by the platform to within a relative deviation of $5 \times 10^{-7}$ in every run, for lags up to 80% of the run. Without subtracting the centre-of-mass motion, the MSD at a lag of 25 ns would be inflated by a median factor of 1.2 and by up to a factor of 34 (more than 10% in 63% of the runs); the largest factors occur in the slowest systems, where a drift of the whole system by a few nanometres is comparable to the displacement of the lithium ions.

## Sensitivity of the regime census and of the lag dependence

Table S2: Sensitivity of the regime census to the lag window used to fit the effective exponent. The main text uses 0.2 to 0.6.

| | 5 ns runs | | | 50 ns runs | | |
|---|---|---|---|---|---|---|
| Fit window (fraction of run) | Median $\beta$ | $\beta < 0.7$ | $\beta \geq 0.9$ | Median $\beta$ | $\beta < 0.7$ | $\beta \geq 0.9$ |
| 0.1 to 0.5 | 0.38 | 99.7% | 0.0% | 0.60 | 81.2% | 0.0% |
| 0.2 to 0.6 | 0.43 | 97.8% | 0.03% | 0.65 | 69.2% | 3.8% |
| 0.3 to 0.7 | 0.48 | 94.9% | 0.07% | 0.69 | 54.9% | 5.3% |
| 0.4 to 0.8 | 0.51 | 87.4% | 0.75% | 0.73 | 45.9% | 14.3% |

Table S3: Sensitivity of the lag dependence in the 133 long runs to the reference lag $\tau_{\mathrm{ref}}$. The main text uses 25 ns.

| | Median $D_{\mathrm{app}}(\tau)/D_{\mathrm{app}}(\tau_{\mathrm{ref}})$ | | | Spearman $\rho$ | | |
|---|---|---|---|---|---|---|
| $\tau_{\mathrm{ref}}$ (ns) | $\tau = 1$ | $\tau = 5$ | $\tau = 10$ | $\tau = 1$ | $\tau = 5$ | $\tau = 10$ |
| 15 | 4.43 | 1.65 | 1.18 | 0.95 | 0.98 | 0.99 |
| 20 | 4.93 | 1.83 | 1.30 | 0.93 | 0.97 | 0.99 |
| 25 | 5.23 | 1.95 | 1.40 | 0.92 | 0.96 | 0.98 |
| 30 | 5.56 | 2.05 | 1.47 | 0.92 | 0.96 | 0.98 |
| 35 | 5.76 | 2.14 | 1.53 | 0.92 | 0.96 | 0.97 |

Table S4: Apparent diffusivity and ranking as a function of the observed lag in the long runs, relative to a lag of 25 ns (curve-based).

| Lag (ns) | Median ratio | 10th to 90th percentile | Within 25% | Spearman $\rho$ |
|---|---|---|---|---|
| 0.5 | 8.86 | 4.70 to 12.47 | 0.0% | 0.91 |
| 1 | 5.23 | 3.12 to 7.26 | 0.0% | 0.92 |
| 2 | 3.34 | 2.15 to 4.32 | 0.0% | 0.94 |
| 5 | 1.95 | 1.47 to 2.40 | 0.8% | 0.96 |
| 10 | 1.40 | 1.18 to 1.63 | 16.5% | 0.98 |
| 15 | 1.19 | 1.08 to 1.31 | 73.7% | 0.99 |
| 20 | 1.07 | 1.03 to 1.12 | 100.0% | 1.00 |

# Run-to-run reproducibility

Table S5: Reproducibility of a single 5 ns run, from 94 systems simulated twice from different initial structures. Intervals by bootstrap over systems.

| Estimator | $\sigma_{\mathrm{run}}$ (log) | Typical factor | Rank corr. between runs | Reliability $r$ (95% CI) | ICC(1) (95% CI) |
|---|---|---|---|---|---|
| End-point | 0.35 | 1.43 | 0.48 | 0.47 (0.31 to 0.63) | 0.53 (0.33 to 0.66) |
| Origin-averaged, lag $T/2$ | 0.28 | 1.32 | 0.56 | 0.58 (0.45 to 0.70) | 0.62 (0.45 to 0.74) |

Two runs of the same system differ by more than a factor of 1.5 in 38% of the replicated systems. The replicated subset is not range-restricted (spread of $\ln D_{\mathrm{app}}$ 0.44 against 0.49 in the whole screening set). The effective exponent of an individual 5 ns run is not reproducible (noise 0.15, rank correlation between runs $-0.07$): it characterises the population, not a single system.

# End-point estimator

The end-point value of the 5 ns runs deviates from the power law fitted over 20% to 60% of the same curve with a standard deviation of 0.19 in log units, a typical factor of 1.21 (Fig. S1); 5.4% of the systems deviate by more than a factor of 1.5, and the rank correlation between the end-point estimate and the origin-averaged estimate at half the run length is 0.92. This deviation combines the single-origin noise of the end point with any departure of the curve from a power law; the replicated runs (Table S5) give the direct comparison of the

two estimators.

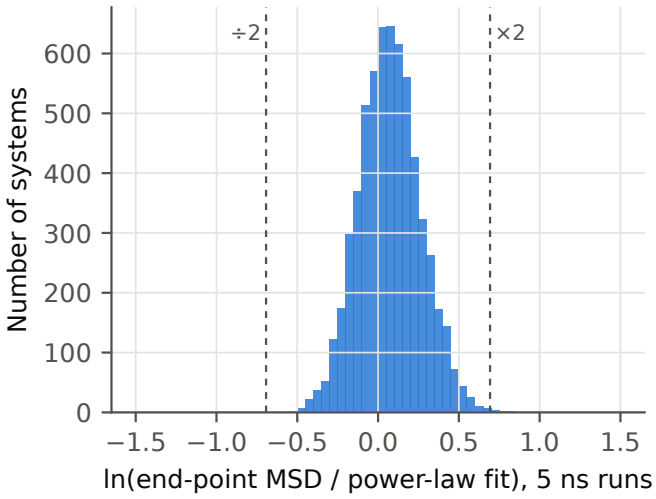



Figure S1: Deviation of the end-point MSD of the 5 ns runs from the power law fitted over lags between 20% and 60% of the run. Dashed lines mark a factor of two.

## Paired short and long runs

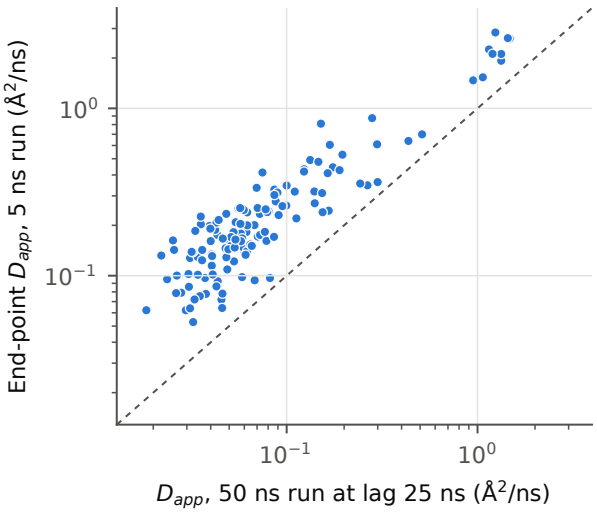



Figure S2: End-point apparent diffusivity from the 5 ns run against the origin-averaged apparent diffusivity at a lag of 25 ns from the 50 ns run of the same system, which starts from the same initial structure (133 systems). The dashed line is the identity.

# Prefix-based estimators and stopping rules

Table S6: Estimates from a prefix of length $T$ relative to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ of the full 50 ns run: median ratio and typical multiplicative error (133 runs).

| | End-point | | Half-lag | | Linear fit | | Power-law extrapolation | |
|---|---|---|---|---|---|---|---|---|
| $T$ (ns) | Ratio | Error | Ratio | Error | Ratio | Error | Ratio | Error |
| 1 | 6.56 | ×6.56 | 10.34 | ×10.34 | 3.22 | ×3.27 | 0.53 | ×1.87 |
| 2 | 4.43 | ×4.43 | 6.39 | ×6.39 | 2.83 | ×2.85 | 0.72 | ×1.50 |
| 5 | 2.83 | ×2.83 | 4.05 | ×4.05 | 2.02 | ×2.02 | 0.95 | ×1.37 |
| 10 | 2.02 | ×2.02 | 2.55 | ×2.55 | 1.48 | ×1.51 | 1.07 | ×1.26 |
| 20 | 1.43 | ×1.45 | 1.78 | ×1.78 | 1.16 | ×1.24 | 1.11 | ×1.20 |
| 30 | 1.20 | ×1.23 | 1.38 | ×1.38 | 0.95 | ×1.20 | 1.05 | ×1.13 |
| 50 | 0.89 | ×1.13 | 1.00 | ×1.00 | 0.73 | ×1.37 | 0.97 | ×1.03 |

The regime rules rarely fire: the effective exponent of a prefix reaches 0.9 before 50 ns in 3% of the runs and 0.8 in 19%, so they save almost no simulation, and when the $\beta \geq 0.8$ rule fires the error of the runs that stop is 1.39. The precision rule with a 20% tolerance stops early in 80% of the runs but with a typical error of 2.22 for those runs: a small bootstrap CV measures the precision of a lag-dependent quantity, not its distance from the long-lag

Table S7: Sensitivity of the prefix-based comparison to the reference (133 long runs). Each cell gives the typical multiplicative error and, in parentheses, the median ratio of the estimate to the reference. The power law is extrapolated to the quantity that defines each reference: the apparent diffusivity at that lag, or the mean slope of $A\tau^{\beta}$ over 20 to 40 ns.

| Reference (from the 50 ns run) | End-point, 5 ns | Linear fit, 5 ns | Power law, 5 ns | Power law, 10 ns |
|---|---|---|---|---|
| $D_{app}$ at 15 ns | ×2.37 (2.37) | ×1.80 (1.73) | ×1.33 (1.06) | ×1.22 (1.16) |
| $D_{app}$ at 25 ns (main) | ×2.83 (2.83) | ×2.02 (2.02) | ×1.37 (0.95) | ×1.26 (1.07) |
| $D_{app}$ at 35 ns | ×3.15 (3.15) | ×2.23 (2.23) | ×1.39 (0.86) | ×1.32 (1.00) |
| $D_{app}$ at 40 ns | ×3.27 (3.27) | ×2.33 (2.33) | ×1.45 (0.83) | ×1.30 (0.95) |
| Slope of MSD, 20 to 40 ns | ×4.01 (4.01) | ×3.05 (3.05) | ×1.99 (0.54) | ×1.52 (0.68) |

Table S8: Stopping rules evaluated on the long runs (133 runs). Runs that do not meet a rule before 50 ns are simulated for the full length and evaluated there. Mean time and the first error column are averaged over all runs; the last column restricts the error to the runs that stop early.

| Rule | Estimator | Stop before 50 ns | Mean time (ns) | Error, all runs | Error, runs that stop |
|---|---|---|---|---|---|
| Regime: $\beta \geq 0.9$ | Linear fit | 3% | 49.3 | ×1.37 | ×1.38 |
| Regime: $\beta \geq 0.8$ | Linear fit | 19% | 44.6 | ×1.39 | ×1.39 |
| Precision: CV < 10% | Linear fit | 2% | 49.3 | ×1.37 | ×1.36 |
| Precision: CV < 20% | Linear fit | 80% | 15.6 | ×1.79 | ×2.22 |
| Precision: CV < 10% | Power law | 24% | 43.4 | ×1.04 | ×1.13 |
| Fixed 5 ns | End-point | n/a | 5.0 | ×2.83 | n/a |
| Fixed 5 ns | Linear fit | n/a | 5.0 | ×2.02 | n/a |
| Fixed 10 ns | Linear fit | n/a | 10.0 | ×1.51 | n/a |
| Fixed 20 ns | Linear fit | n/a | 20.0 | ×1.24 | n/a |
| Fixed 5 ns | Power law | n/a | 5.0 | ×1.37 | n/a |
| Fixed 10 ns | Power law | n/a | 10.0 | ×1.26 | n/a |
| Fixed 20 ns | Power law | n/a | 20.0 | ×1.20 | n/a |

value.

Table S9: Settings chosen on half of the systems and evaluated on the other half (median over 500 random partitions): mean simulated time, typical error, and percentage of partitions in which the held-out half meets the target; n/a means that no setting met the target on the half used for the choice.

| | Target error ×1.5 | | | Target error ×1.3 | | |
|---|---|---|---|---|---|---|
| Rule | Time (ns) | Error | Target met | Time (ns) | Error | Target met |
| Fixed length, end-point | 20.0 | ×1.45 | 97% | 30.0 | ×1.23 | 84% |
| Fixed length, linear fit | 12.0 | ×1.47 | 61% | 20.0 | ×1.24 | 96% |
| Fixed length, power law | 2.0 | ×1.50 | 59% | 7.5 | ×1.31 | 45% |
| Regime rule, linear fit | 35.2 | ×1.48 | 64% | n/a | n/a | 0% |
| Precision rule, linear fit | 37.0 | ×1.47 | 67% | 50.0 | ×1.40 | 0% |
| Precision rule, power law | 11.9 | ×1.43 | 76% | 20.9 | ×1.18 | 93% |

# Robustness of the adaptive allocation

Table S10: Mean number of 5 ns runs per candidate that boundary-focused allocation needs to match the recall of uniform replicates, and the corresponding saving (end-point label, Gaussian system values, 200 simulated campaigns of 6,137 candidates).

| | Uniform replicates matched | | |
|---|---|---|---|
| Target | 2 runs | 3 runs | 4 runs |
| Top 1% | 1.1 (×1.7) | 1.2 (×2.5) | 1.2 (×3.2) |
| Top 5% | 1.2 (×1.7) | 1.3 (×2.2) | 1.5 (×2.7) |
| Top 10% | 1.2 (×1.6) | 1.5 (×2.0) | 1.8 (×2.2) |

The saving for the top 10% against four uniform replicates is ×2.2 when the system values follow the empirical shape of the 6,137 short runs instead of a normal distribution, and ×2.4 with the origin-averaged label. Using the mobility-dependent noise in the allocation does not help: the recall of the top 10% with four runs per candidate is 84% with it and 85% with a constant noise. Successive halving with restarted rounds performs similarly to boundary-focused allocation. With the end-point label and 1.25 runs per candidate it recovers 74% of the top 1%, against 68%; with the origin-averaged label and four runs per candidate it recovers 84%, against 89%.

## Check of the adaptive allocation on real runs

Table S11: Check of the adaptive allocation on the replicated systems. Calibration: 94 systems, predictions in both directions (first run → second and second → first), noise parameters estimated leaving out the system evaluated. Campaign: 92 systems with a long run; every system receives its first run and 46 receive a second run, chosen at the boundary or at random (mean of 2000 draws); recall of the top set according to the long run. Interval: bootstrap over systems of the difference between boundary and random placement.

| | Top 10% | Top 20% |
|---|---|---|
| Brier improvement over the base rate | 36% | 16% |
| Calibration slope (95% CI) | 1.10 (0.65 to 2.28) | 0.63 (0.34 to 1.35) |
| Systems in the top set, $k$ | 9 | 18 |
| Recall, first runs only | 67% | 72% |
| Recall, second runs at random | 71% | 71% |
| Recall, second runs at the boundary | 78% | 72% |
| Recall, second runs for all | 78% | 72% |
| Boundary − random (95% CI, points) | −17 to +17 | −15 to +15 |

The long runs start from the same structure as the first runs, so they favour the first run and understate the value of any second run from a new structure; this is why no strategy improves on the first run alone for the top 20%, and why the check is conservative.